



ASCHAM SCHOOL
2002 TRIAL HSC EXAMINATION

MATHEMATICS EXTENSION 1
FORM VI

General Instructions:

- Reading Time: 5 minutes
- Working Time: 2 hours
- Write using blue or black pen
- Approved calculators and templates may be used.
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question.

Collection:

- Start each question in a new answer book
- Write your name and teacher's name on each book.
- If you use a second book, place it inside the first.

Total Marks : 84

- Attempt Questions 1 – 7
- All questions are of equal value.

Question 1 Start a new answer book

- a) Express $\frac{5\pi}{12}$ radians as degrees
[1]
- b) Find a primitive of e^{-2x} [1]
- c) Use the table of standard integrals to find the exact value of

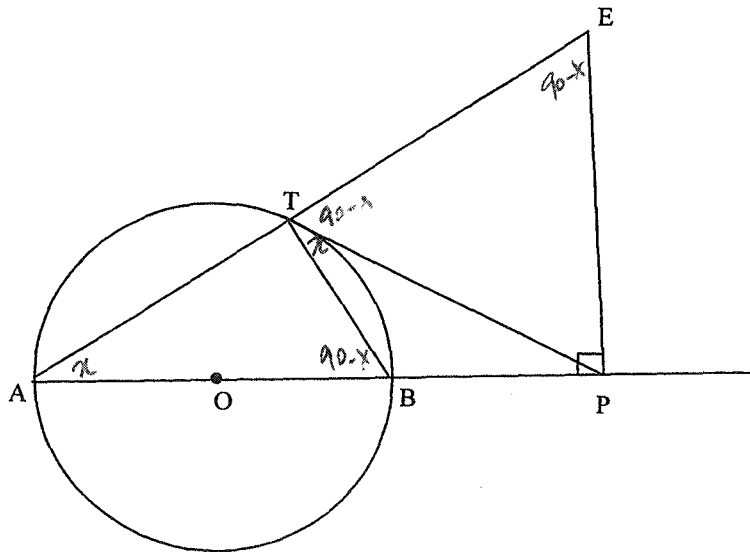
$$\int_0^4 \frac{dx}{\sqrt{x^2 + 4}}$$
 [2]
- d) If α, β and γ are roots of the equation $6x^3 + 7x^2 - x - 2 = 0$, find the value of

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$
 [3]
- e) Find the domain and range of $f(x) = 4 \sin^{-1} \frac{x}{3}$, and sketch the graph of $f(x)$.[3]
- f) Find $\frac{d}{dx} e^{\cos x}$ [2]

Question 2 Start a new answer book

- a) Find (i) $\int \frac{x}{4 + x^2} dx$ [1]
(ii) $\int_{\frac{\pi}{8}}^{\frac{\pi}{6}} \tan^2 2x dx$ [3]
- b) If $\sum_{k=4}^{\infty} 2r^{k-3} = 10$, find r if r exists. [3]

c)



Make a large neat copy of the diagram in your answer book.

AB is the diameter of the circle with centre O. TP is a tangent to the circle at T. $EP \perp AP$. Prove:

- (i) TBPE is a cyclic quadrilateral [2]
- (ii) $PT = PE$. [3]

Question 3 **Start a new answer book**

a) Solve : $\frac{2x}{5-x} \geq 1$ [3]

b) Prove by mathematical induction that $6^n - 1$ is divisible by 5 for all positive integers.

[5]

c) By substituting $t = \tan \frac{x}{2}$, find the solutions to the equation:
 $3 \sin x + 4 \cos x = 5$ for $0^\circ \leq x \leq 360^\circ$,
 giving your answers correct to the nearest degree.
 [4]

Question 4

Start a new answer book

- a) Using the substitution $u = x^3 + 1$, evaluate $\int_{-1}^1 x^2(x^3 + 1) dx$ [2]
- b) (i) Factorise : $x^3 - 3x + 2$
 (ii) Hence draw a neat sketch of the polynomial $y = x^3 - 3x + 2$ without the use of calculus, showing all intercepts with the co-ordinate axes.
 (iii) Hence solve the inequality $x^3 - 3x + 2 > 0$ [4]
- c) Find the value of $\sin\left(2 \sin^{-1} \frac{2}{3}\right)$ in exact form [3]
- d) i) Show that the equation $f(x) = x^3 - 8x + 8$ has a zero between -3 and -4 .
 ii) Taking $x = -3.5$ as a first approximation of the solution of the equation $f(x) = 0$, use Newton's method once to find a closer approximation, giving your answer to 2 decimal places [3]

Question 5

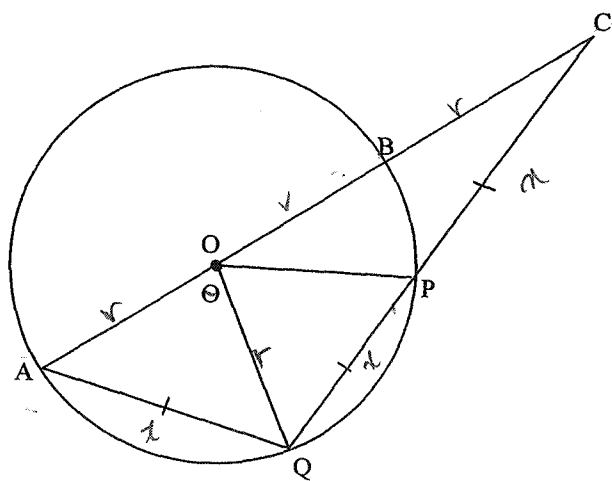
Start a new answer book

- a) A bug is oscillating in simple harmonic motion such that its displacement x metres from a fixed point O at time t seconds is given by the equation $\ddot{x} = -4x$. When $t = 0$, $v = 2$ m/s and $x = 5$.
 (i) Show that $x = a \cos(2t + \beta)$ is a solution for this equation, where a and β are constants.
 (ii) Find the period of the motion.
 (iii) Show that the amplitude of the oscillation is $\sqrt{26}$.
 (iv) What is the maximum speed of the bug? [5]
- b) (i) Prove that $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = \ddot{x}$ [2]
 (ii) The acceleration of a creature is given by $\ddot{x} = -\frac{1}{2}u^2 e^{-x}$ where x is the displacement from the origin, and u is the initial velocity at the origin. Given that $u = 2$ m/s:
 (α) Show that $v^2 = 4e^{-x}$
 (β) Explain why $v > 0$, and find x in terms of t .
 (γ) Describe the subsequent motion of the creature as $t \rightarrow \infty$. [5]

Question 6 Start a new answer book

- a) A ladder is slipping down a vertical wall. The ladder is 4 metres long. The top of the ladder is slipping down at a rate of 3 m/s. How fast is the bottom of the ladder moving along the ground when the bottom is 2 metres away from the foot of the wall?
[4]
- b) The point $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$. The focus S is the point $(0, a)$. The tangent at P meets the y-axis at Q .
- (i) Find the equation of the tangent at P and the co-ordinates of Q .
 - (ii) Prove that $SP = SQ$
 - (iii) Hence show that $\angle PSQ + 2\angle SQP = 180^\circ$ [4]
- c) In a town in Mathsland, a 'flu epidemic is spreading at a rate proportional to the population that have it, such that it is predicted that the number of people who have the disease will double in 3 weeks, i.e. $\frac{dA}{dt} = kA$, where A is the number of people with 'flu in time t weeks.
- (i) Show that $A = A_0 e^{kt}$, where A_0 is the initial number of people with 'flu, satisfies the above differential equation.
 - (ii) Find k in exact form
 - (iii) In the neighbouring town with a population of 20,000, three people have the 'flu. How many weeks (to the nearest week) will it take for the whole population to contract the disease?
[4]

Question 7 Start a new book



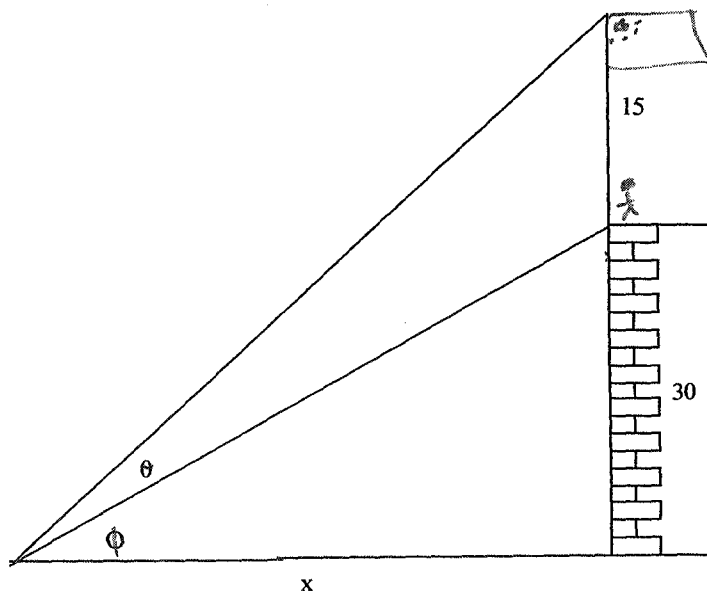
- a) *Make a large neat copy of the diagram in your answer book.*

AB is the diameter of the circle with centre O and radius r . $BC = r$,
 $AQ = QP = PC$, and $\angle AOQ = \theta$

- (i) Prove that $\cos \theta = \frac{1}{4}$ (Hint: use the cosine rule in triangles AQO
 and QOC) [5]

- (ii) Hence prove that $QC = r\sqrt{6}$ [2]

- b) A 15 metre high flagpole stands on top of a building which is 30 m high. The flagpole subtends an angle of θ degrees to a point x metres from the foot of the building, and the building subtends an angle of ϕ degrees to the



same point.

- i) Show that $\theta = \tan^{-1} \left(\frac{15x}{x^2 + 1350} \right)$
- ii) Hence find the value of x which will make θ a maximum. [6]

End of Examination

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

ASCHAM

SOLUTIONS: EXTENSION 1 TRIAL EXAM 2002

DKG

1) a) $\frac{7\pi}{12} = \boxed{75^\circ}$

b) $\int e^{-2x} dx = \boxed{-\frac{1}{2} e^{-2x} + c}$

c) $\int_0^4 \frac{dx}{\sqrt{x^2+4}} = \left[\log_e(x + \sqrt{x^2+4}) \right]_0^4$
 $= \log_e(4 + \sqrt{20}) - \log_e(0 + 2)$
 $= \log_e(4 + 2\sqrt{5}) - \log_e 2$
 $= \boxed{\log_e(2 + \sqrt{5})}$ //

d) $6x^3 + 7x^2 - x - 2 = 0$

$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$
 $= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$

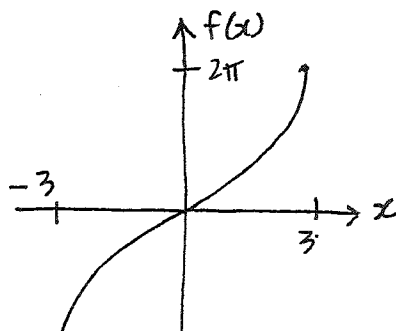
$= \frac{-\frac{1}{6}}{\frac{1}{3}}$
 $= \boxed{-\frac{1}{2}}$ //

$\alpha\beta\gamma = -\frac{d}{a} = \frac{1}{3}$
 $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = -\frac{1}{6}$

e) $f(x) = 4 \sin^{-1} \frac{x}{3}$
 let $y = 4 \sin^{-1} \frac{x}{3}$

$\therefore \frac{y}{4} = \sin^{-1} \frac{x}{3}$

range: $-2\pi \leq y \leq 2\pi$
 dom: $-1 \leq \frac{x}{3} \leq 1$



2) f) $\frac{d}{dx} e^{\cos x} = \boxed{-\sin x \cdot e^{\cos x}}$

a) i) $\int \frac{x}{4+x^2} dx = \frac{1}{2} \int \frac{2x}{4+x^2} dx = \boxed{\frac{1}{2} \log(4+x^2)}$

ii) $\int_{\frac{\pi}{8}}^{\frac{\pi}{6}} \tan^2 2x dx = \int_{\frac{\pi}{8}}^{\frac{\pi}{6}} (\sec^2 2x - 1) dx$
 $= \left[\frac{1}{2} \tan 2x - x \right]_{\frac{\pi}{8}}^{\frac{\pi}{6}}$
 $= \frac{1}{2} \tan \frac{\pi}{3} - \frac{\pi}{6} - \left(\frac{1}{2} \tan \frac{\pi}{4} - \frac{\pi}{8} \right)$
 $= \frac{1}{2} \sqrt{3} - \frac{\pi}{6} - \frac{1}{2} + \frac{\pi}{8}$
 $= \frac{1}{2} \sqrt{3} - \frac{1}{2} - \frac{\pi}{24}$
 $= \boxed{\frac{1}{24} (12\sqrt{3} - 12 - \pi)}$

b) $\sum_{k=4}^{\infty} ar^{k-3} = ar^1 + ar^2 + ar^3 + \dots$

so $10 = a(r + r^2 + r^3 + \dots)$

$\therefore 5 = r + r^2 + r^3 + \dots$

$\therefore 5 = \frac{r}{1-r}$

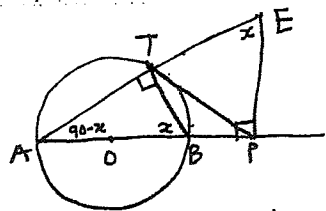
$a=r \quad r=r$
 $S_{\infty} = \frac{a}{1-r}$

$\therefore 5 - 5r = r$

$6r = 5$

$r = \frac{5}{6}$

for S_{∞} $-1 < r < 1$



(i) $\angle ATB = 90$ ($\angle$ in semi $\odot$)

$$= \angle EPB$$

$\therefore$ TPBE is cyclic Q.E.D.

(ext $\angle$ = int opp $\angle$ of quad).

(ii) let $\angle E = x$

$$\therefore \angle TAB = 90 - x \quad (\angle \text{'s of } \triangle EPA)$$

$$\therefore \angle PTB = 90 - x \quad (\angle \text{ in alt seg.})$$

$$\therefore \angle TEP = x \quad (\angle \text{'s on str line})$$

$$\therefore TP = TE \quad \text{Q.E.D. (sides opp } \angle \text{'s)}$$

$$\frac{2x}{5-x} \geq 1$$

$$x \neq 5$$

$$x(5-x)^2$$

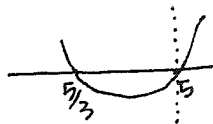
$$2x(5-x) \geq (5-x)^2$$

$$10x - 2x^2 \geq 25 - 10x + x^2$$

$$3x^2 - 20x + 25 \leq 0$$

$$(3x-5)(x-5) \leq 0$$

$$\text{Soln: } \frac{5}{3} \leq x < 5$$



let $P(n) = 6^n - 1$
3b) when $n=1$ $6^n - 1 = 6 - 1 = 5$

which is divisible by 5

let us assume $\exists k$ such that
 $6^k - 1 = 5m$ for some $m \in \mathbb{Z}$

we want to show that $P(k+1)$ is divis by 5

i.e $6^{k+1} - 1$ is divisible by 5.

$$\text{now } 6^{k+1} - 1 = 6^k \cdot 6 - 1$$

$$\text{from } \square: 6^k = 5m + 1$$

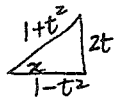
$$\begin{aligned} \text{so } 6^{k+1} - 1 &= (5m + 1)6 - 1 \\ &= 6 \cdot 5m + 6 - 1 \\ &= 6 \cdot 5m + 5 \\ &= 5(6m + 1) \end{aligned}$$

which is divisible by 5.

so by the process of mathematical induction, the statement $P(n)$ is true for all $n \in \mathbb{Z}^+$

$$3 \sin x + 4 \cos x = 5$$

let $t = \tan \frac{1}{2}x$



$$\frac{6t}{1+t^2} + \frac{4(1-t^2)}{1-t^2} = 5$$

$$6t + 4 + 4t^2 = 5 + 5t^2$$

$$4t^2 - 6t + 1 = 0$$

$$(3t-1)(3t-1) = 0$$

$$\therefore t = \frac{1}{3} \quad \text{for } 0^\circ \leq x \leq 180^\circ$$

$$\text{so } \tan \frac{1}{2}x = \frac{1}{3}$$

$$\frac{1}{2}x = 18^\circ 26'$$

$$x = 36^\circ 52'$$

$$\text{testing } x = 180^\circ:$$

$$\text{LHS} = 3 \times 0 + 4 \times -1$$

$$= -4$$

$$\neq \text{RHS}$$

$$\text{so soln is } x = 36^\circ 52' = 37^\circ \text{ (to n degree)}$$

$$\int_{-1}^1 x^2 (x^3+1) dx$$

$$= \frac{1}{3} \int_0^2 u du$$

$$= \frac{1}{3} \left[\frac{1}{2} u^2 \right]_0^2$$

$$= \frac{1}{6} (4 - 0)$$

$$= \boxed{\frac{2}{3}}$$

$$u = x^3 + 1$$

$$du = 3x^2 dx$$

$$\therefore x^2 dx = \frac{1}{3} du$$

when $x=1$, $u=2$
 $x=-1$, $u=0$

$$4b) i) f(x) = x^3 - 3x + 2$$

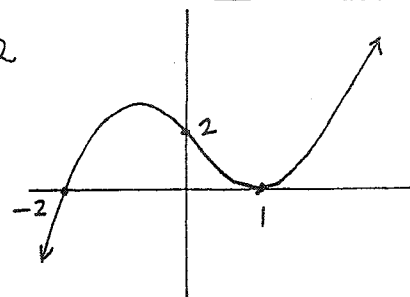
$$f(1) = 1 - 3 + 2 = 0$$

so $(x-1)$ is a factor

$$\begin{array}{r} x^2 + x - 2 \\ x-1 \overline{) x^3 - 3x + 2} \\ \underline{x^3 - x^2} \\ x^2 - 3x \\ \underline{x^2 - x} \\ -2x + 2 \\ \underline{-2x + 2} \\ 0 \end{array}$$

$$\text{so } f(x) = (x-1)(x+2)(x-1)$$

$$ii) y\text{-int} = 2$$



$$iii) x^3 - 3x + 2 > 0$$

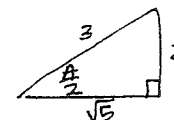
for $-2 < x < 1$ and $x > 1$ (or $x > 1$)

$$c) \sin(2 \sin^{-1} \frac{2}{3})$$

$$\text{let } A = \sin^{-1} \frac{2}{3}$$

$$\therefore A = \sin^{-1} \frac{2}{3}$$

$$\therefore \sin A = \frac{2}{3}$$



$$\text{hence } \sin 2A = 2 \sin A \cos A$$

$$\begin{aligned}
 \text{i)} \quad f(x) &= x^3 - 8x + 8 \\
 f(-3) &= (-3)^3 - 8(-3) + 8 \\
 &= 5 \\
 f(-4) &= (-4)^3 - 8(-4) + 8 \\
 &= -24
 \end{aligned}$$

$\therefore f(x)$ has a zero b/w -3 and -4

$$\begin{aligned}
 \text{i)} \quad x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \quad f'(x) = 3x^2 - 8 \\
 \therefore x_1 &= -3.5 - \frac{(-3.5)^3 - 8(-3.5) + 8}{3(-3.5)^2 - 8} \\
 &= -3.26086 \dots
 \end{aligned}$$

$$\doteq -3.26$$

so $x = -3.26$ is an approx for $f(x) = 0$.

7]

$$\begin{aligned}
 \text{i)} \quad \text{i)} \quad x &= a \cos(2t + \beta) \\
 \therefore v &= -2a \sin(2t + \beta) \\
 \ddot{x} &= -4a \cos(2t + \beta) \\
 \text{so } \ddot{x} &= -4x \text{ and satisfies eqn.}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad T &= \frac{2\pi}{n} \\
 T &= \frac{2\pi}{2} \\
 &= \pi
 \end{aligned}$$

$$\begin{aligned}
 \text{iii)} \quad \text{when } t=0, x=5 \quad | \quad \text{when } t=0, v= \\
 \therefore 5 = a \cos \beta \quad | \quad \therefore 2 = -2a \sin \beta \\
 \text{squaring and adding: } -1 = a \sin \beta \\
 26 = a^2 \\
 \therefore a = \sqrt{26} \text{ since } a > 0 \\
 \text{so amp} = \sqrt{26}
 \end{aligned}$$

$$\begin{aligned}
 \text{iv)} \quad v &= -2\sqrt{26} \sin(2t + \beta) \\
 \text{so max speed is } 2\sqrt{26} \text{ m/s} \\
 (\text{since } -2\sqrt{26} \sin(2t + \beta) \text{ has max})
 \end{aligned}$$

$$\text{5 b) i) to prove: } \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \ddot{x}$$

$$\begin{aligned}
 \text{proof: } \frac{d}{dx} \left(\frac{1}{2} v^2 \right) &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \cdot \frac{dv}{dx} \\
 &= v \cdot \frac{dv}{dx} \\
 &= \frac{dx}{dt} \cdot \frac{dv}{dx} \\
 &= \frac{dv}{dt} \\
 &= \frac{d^2 x}{dt^2} \\
 &= \ddot{x} \quad \text{Q.E.D.}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad \ddot{x} &= -\frac{1}{2} u^2 e^{-x} \\
 u &= 2 \text{ m/s}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \ddot{x} &= -\frac{1}{2} \times 4 e^{-x} \\
 \ddot{x} &= -2e^{-x} \\
 \therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) &= -2e^{-x}
 \end{aligned}$$

$$\therefore \frac{1}{2} v^2 = 2e^{-x} + \frac{1}{2} C_1$$

$$\therefore v^2 = 4e^{-x} + C_1$$

when $x=0$, $v=2$

$$\therefore 4 = 4 + C_1$$

$$\text{so } C_1 = 0$$

$$\therefore \underline{v^2 = 4e^{-x}} \quad \text{P.E.D.}$$

β) $4e^{-x} > 0$ for all x
 so v^2 does not change sign.
 since $v=2$ at $x=0$, it
 remains positive.

$$\therefore \boxed{v = 2e^{-\frac{x}{2}}}$$

$$\frac{dx}{dt} = \frac{2}{e^{x/2}}$$

$$\frac{dt}{dx} = \frac{e^{x/2}}{2}$$

$$\therefore t = \frac{1}{2} \cdot 2e^{x/2} + C_2$$

$$t = e^{x/2} + C_2$$

at $x=0$, $t=0$

$$0 = 1 + C_2$$

$$\therefore C_2 = -1$$

$$\therefore t = e^{x/2} - 1$$

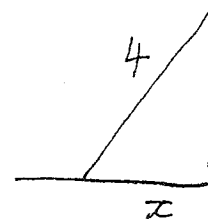
$$\therefore e^{x/2} = t + 1$$

$$\log(t+1) = \frac{x}{2}$$

$$\therefore \boxed{x = 2 \log_e(t+1)}$$

γ) as $t \rightarrow \infty$, $x \rightarrow \infty$ so bug displacement increases it

$\boxed{6}$ a) $\frac{dy}{dt} = -3$ m/s
 we want $\frac{dx}{dt}$ when $x=2$.



by pythag:

$$x^2 = 16 - y^2$$

$$\therefore x = \sqrt{16 - y^2} = (16 - y^2)^{\frac{1}{2}}$$

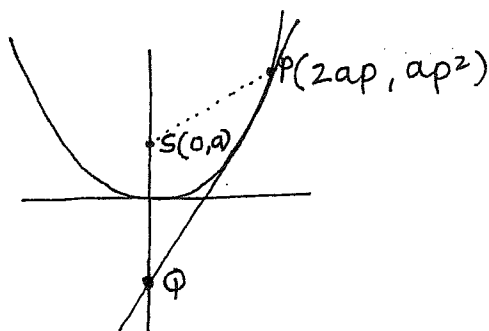
$$\therefore \frac{dx}{dy} = \frac{1}{2}(16 - y^2)^{-\frac{1}{2}} \cdot -2y$$

$$\frac{dx}{dy} = -\frac{y}{\sqrt{16 - y^2}}$$

also, when $x=2$, $y^2 = 16 - 4$
 $= 12$
 $\therefore y = 2\sqrt{3}$

$$\begin{aligned} \text{now } \frac{dx}{dt} &= \frac{dx}{dy} \times \frac{dy}{dt} \\ &= \frac{-y}{\sqrt{16 - y^2}} \times -3 \\ &= \frac{-2\sqrt{3}}{\sqrt{16 - 12}} \times -3 \\ &= 3\sqrt{3} \end{aligned}$$

so foot is sliding away at rate
of $3\sqrt{3}$ m/s #



1. grad of tang = p

$$\therefore \text{eqn is } y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$

when $x = 0$, $y = -ap^2$

$$\therefore Q = (0, -ap^2)$$

i) $\text{dist}^2 SQ = (a + ap^2)^2$ #

$$\text{dist}^2 SP = (2ap)^2 + (ap^2 - a)^2$$

$$= 4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2$$

$$= a^2 + 2a^2p^2 + a^2p^4$$

$$= (a + ap^2)^2$$

$\therefore SQ = SP$ Q.E.D. #

ii) $\angle SQP = \angle SPQ$ ($SQ = SP$)

$\therefore \angle PSQ + 2\angle SQP = 180^\circ$ ($\angle$ sum of $\triangle PSQ$) #

6 c) i) sub $A = A_0 e^{kt}$ into $\frac{dA}{dt} = kA$

$$\text{LHS} = \frac{d}{dt}(A_0 e^{kt})$$

$$= A_0 \cdot k e^{kt}$$

$$\text{RHS} = k(A_0 e^{kt})$$

$$= A_0 k e^{kt}$$

$= \text{LHS}$ # Q.E.D.

ii) when $t = 3$, $A = 2A_0$

$$\therefore 2A_0 = A_0 e^{3k}$$

$$e^{3k} = 2$$

$$k = \frac{1}{3} \log_e 2$$

iii) $A = A_0 e^{kt}$ where $k = \frac{1}{3} \ln 2$

when $A_0 = 3$, $A = 20,000$

$$\therefore 20,000 = 3 e^{kt}$$

$$e^{kt} = \frac{20000}{3}$$

$$\therefore t = \frac{1}{k} \log_e \frac{20000}{3}$$

$$= 38.108249... \text{ weeks}$$

$$\therefore t \approx 39$$

it will take 39 complete weeks #

let $AQ = x$

i) in $\triangle APO$

$$x^2 = 2r^2 - 2r^2 \cos \theta \quad \text{--- [1]}$$

in $\triangle POC$

$$QC^2 = r^2 + 4r^2 - 2r \cdot 2r \cos (180 - \theta)$$

$$4x^2 = 5r^2 + 4r^2 \cos \theta$$

$$\therefore x^2 = \frac{5}{4}r^2 + r^2 \cos \theta \quad \text{--- [2]}$$

$$\text{so } 2r^2 - 2r^2 \cos \theta = \frac{1}{5}r^2 + r^2 \cos \theta$$

$$\text{[2]} \quad 2 - 2 \cos \theta = \frac{5}{4} + \cos \theta$$

$$3 \cos \theta = \frac{3}{4}$$

$$\cos \theta = \frac{1}{4} \quad \text{Q.E.D.} \quad \#$$

i) from [2]

$$QC^2 = 5r^2 + 4r^2 \cos \theta$$

$$QC^2 = 5r^2 + 4r^2 \times \frac{1}{4}$$

$$QC^2 = 6r^2$$

$$QC = r\sqrt{6} \quad \text{Q.E.D.} \quad \#$$

7b) i) $\tan \phi = \frac{30}{x}$
 $\tan(\theta + \phi) = \frac{45}{x}$

$$\tan(\theta + \phi) = \frac{\tan \theta + \tan \phi}{1 - \tan \theta \cdot \tan \phi}$$

$$\frac{45}{x} = \frac{\tan \theta + \frac{30}{x}}{1 - \tan \theta \cdot \frac{30}{x}}$$

$$\frac{45}{x} = \frac{x \tan \theta + 30}{x - 30 \tan \theta}$$

$$\therefore 45x - 1350 \tan \theta = x^2 \tan \theta + 30x$$

$$(x^2 + 1350) \tan \theta = 45x - 30x$$

$$\tan \theta = \frac{15x}{x^2 + 1350}$$

$$\theta = \tan^{-1} \left(\frac{15x}{x^2 + 1350} \right)$$

$$\text{ii) } \frac{d\theta}{dx} = \frac{1}{1 + \left(\frac{15x}{x^2 + 1350} \right)^2} \times \frac{(x^2 + 1350)15 - 15x \cdot 2x}{(x^2 + 1350)^2}$$

$$= \frac{15x^2 + 20250 - 30x^2}{(x^2 + 1350)^2 + (15x)^2}$$

$$= \frac{20250 - 15x^2}{(x^2 + 1350)^2 + (15x)^2}$$

$$\text{max occurs when } \frac{d\theta}{dx} = 0$$

$$\therefore 15x^2 = 20250$$

$$x^2 = 1350$$

$$x = 15\sqrt{6}$$

test:		
x	36	$15\sqrt{6}$
$\frac{d\theta}{dx}$	$\frac{810}{+}$	0